

Lecture 8: Measuring Downside Risk With VaR And CVaR

Phase 2: Risk Management Science

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Nurturing Future Risk Managers and Actuaries

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1. Opening problem: How bad can the worst 5% be?
2. Part A. CDF and quantile: data-first view using ECDF and percentile cutoffs.
3. Part B. Returns and basic risk descriptors: arithmetic return, log return, mean, and volatility.
4. Part C. Value-at-Risk: historical VaR and Normal-approximation VaR.
5. Part D. Conditional VaR / Expected Shortfall: average outcome inside the worst tail.
6. Part E. Investigation and review: technology-heavy application and interpretation practice.

Technology Check: Central Workflow

Data → returns → empirical distribution → quantile cutoff → interpretation sentence → stress-test assumptions.

Cognitive Trigger / Opening Problem

You manage a small portfolio, such as a student investment club fund. You want a **one-number** summary of tomorrow's downside risk.

Investigation / Activity: Guiding Questions

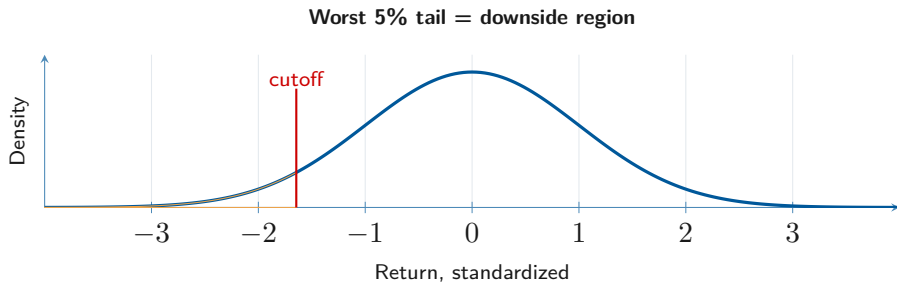
- ▶ What loss level will only be exceeded on 5% of days? This is the idea of **Value-at-Risk**.
- ▶ If we are already in that worst 5%, what is the **average** loss? This is the idea of **CVaR / Expected Shortfall**.
- ▶ What data and what assumptions do we need to answer quickly?

Pause-and-Think

If two portfolios have the same “worst 5% cutoff”, must they be equally dangerous?

Key Definition / Result

Idea: VaR is a **cutoff**, meaning a quantile. CVaR looks **deeper** into the tail.



Key Definition / Result

VaR and CVaR are computed from **data**. We need reliable estimates and, sometimes, a quick distributional approximation.

- ▶ **Law of Large Numbers:** with more data, averages become more stable, so mean and volatility estimates become more reliable.
- ▶ **Central Limit Theorem:** sums and averages of many shocks often look bell-shaped.
- ▶ **Normal approximation:** fast quantiles, or z-values, lead to a quick VaR formula.

Common Mistake / Pitfall

The Law of Large Numbers **does not** say that short runs must “balance out”. That is the gambler’s fallacy. It says long-run averages stabilise.

Part A. CDF And Quantile: Data-First View



Key Definition / Result

The **cumulative distribution function**, or CDF, of a random variable X is $F(x) = P(X \leq x)$. It converts a **value** x into a **probability**.

Working Example: Discrete Case

For a discrete random variable, $F(x)$ is the cumulative sum of probabilities up to x .

Working Example: Continuous Case

For a continuous random variable, $F(x)$ is the cumulative area up to x . We do not need formal calculus in lecture.

Pause-and-Think

Why is the CDF easier to compare across different types of random variables than the probability mass function or density alone?

Worked Example

Let X be the result of a fair six-faced dice. Find $F(x) = P(X \leq x)$.

1. List the probabilities: $P(X = k) = \frac{1}{6}$ for $k = 1, \dots, 6$.
2. For each x , add all probabilities with outcomes $\leq x$.
3. For example, $F(3) = P(X \leq 3) = \frac{3}{6} = \frac{1}{2}$.

Key Definition / Result

Key idea: the CDF is accumulating probability.

Technology Check

Use technology to confirm the CDF idea through simulation and the empirical cumulative distribution function.

Technology Check: R Code Workflow

```
> set.seed(1)
> dice_res <- sample(1:6, 200,
  replace=T)
> emp_cdf <- ecdf(dice_res)
> emp_cdf(3) # approx  $P(X \leq 3)$ 
> plot(emp_cdf)
```

Key Definition / Result

Given data $x_1, \dots, x_n$, the **empirical CDF** is

$$\hat{F}(x) = \frac{\#\{x_i \leq x\}}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x_i \leq x)$$

Investigation / Activity: Interpretation Sentence

R

read $\hat{F}(x)$ as: “What fraction of observations are $\leq x$?”

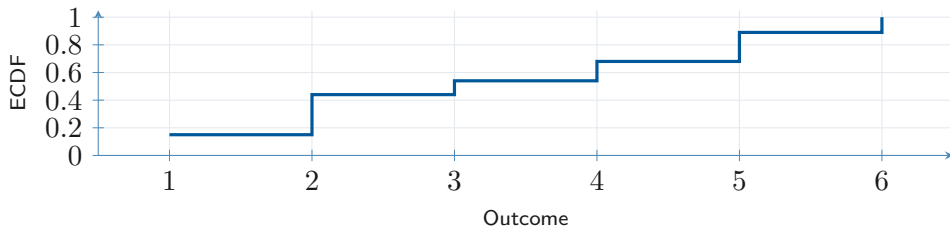
Common Mistake / Pitfall

Do not confuse $\hat{F}(x)$, which is a fraction, with the data values x_i , which may be returns, losses, times, or other numerical observations.

Key Definition / Result

An ECDF is a **step function**. It jumps up by $1/n$ at each observed value.

Example ECDF shape from dice data



Worked Example

Suppose 100 throws give counts 1, 2, 3, 4, 5, 6 with frequencies 15, 29, 10, 14, 21, 11. Compute the empirical probability mass function and ECDF.

1. Empirical PMF: $\hat{p}(k) = \frac{\text{freq. of } k}{100}$.
2. ECDF: $\hat{F}(1) = \hat{p}(1)$, $\hat{F}(2) = \hat{p}(1) + \hat{p}(2)$.
3. Example: $\hat{F}(4) = \frac{15+29+10+14}{100} = 0.68$.

Outcome	Freq.
1	15
2	29
3	10
4	14
5	21
6	11

Key Definition / Result

The p -quantile is the value x such that $P(X \leq x) = p$.

CDF: value $\rightarrow$ probability Quantile: probability $\rightarrow$ cutoff value.

- ▶ Median = 0.50 quantile.
- ▶ “5% quantile” = cutoff so that only 5% are below.

Pause-and-Think

Why is a quantile naturally suited for describing a worst-case threshold?

Worked Example

For a fair dice, what is the 0.833 quantile? What is the median?

1. Compute cumulative probabilities: $F(1) = 1/6$, $F(2) = 2/6$, ..., $F(6) = 1$.
2. Find the smallest x such that $F(x) \geq p$.

Key Definition / Result: Answer

- ▶ For $p = 0.833 \approx 5/6$, the smallest x with $F(x) \geq 5/6$ is $x = 5$.
- ▶ Median ($p = 0.5$): the smallest x with $F(x) \geq 0.5$ is $x = 3$.

Technology Check

Quantiles are built in. Use technology to avoid manual sorting for large datasets.

Key Definition / Result: Excel Concept

- ▶ `=PERCENTILE.INC(range, p)`
- ▶ Example:
`=PERCENTILE.INC(A2:A251, 0.05)`

Technology Check: R Code

```
> quantile(x, probs = 0.05)
> quantile(x, probs = 0.50)
```

Worked Example

A loss X has density “proportional to x ” on $0 < x < 10000$. Find the constant and the 95% quantile.

Deepening Layer: Lecture-Level Approach

We avoid full integration in lecture. Instead, we solve using a known CDF form or technology. In tutorial, derive $F(x) = \left(\frac{x}{10000}\right)^2$ for this model.

Key Definition / Result

If $F(x) = (x/10000)^2$, then the 95% quantile satisfies:

$$\left(\frac{x}{10000}\right)^2 = 0.95 \implies x = 10000\sqrt{0.95}$$

- ▶ **Bernoulli/Binomial:** quantiles are stepwise discrete cutoffs.
- ▶ **Poisson:** quantiles tell us how many events occur in a worst $p\%$ year.
- ▶ **Normal:** quantiles come from z-values, standard Normal tables, or software.

Technology Check

We will use R functions: `qbinom`, `qpois`, `qnorm`.

Worked Example

Suppose protests per year follow $X \sim \text{Poisson}(\lambda = 10)$. Find $P(X \leq 1)$ and the 5% quantile.

1. Compute probability with `ppois(1, lambda=10)`.
2. Compute quantile with `qpois(0.05, lambda=10)`.

Technology Check: R Code

```
> poisson_rate <- 10  
> ppois(1, lambda = poisson_rate)  
> qpois(0.05, lambda = poisson_rate)
```

Part B. Returns And Basic Risk Descriptors

Key Definition / Result

For prices S_{t-1} and S_t , the arithmetic return is $R_t = \frac{S_t - S_{t-1}}{S_{t-1}}$.

The log return is $u_t = \log\left(\frac{S_t}{S_{t-1}}\right) = \log(1 + R_t)$.

1. Compute each arithmetic return from two consecutive prices.
2. Compute each log return using logarithms.
3. Check the approximation: for small returns, $u_t \approx R_t$.

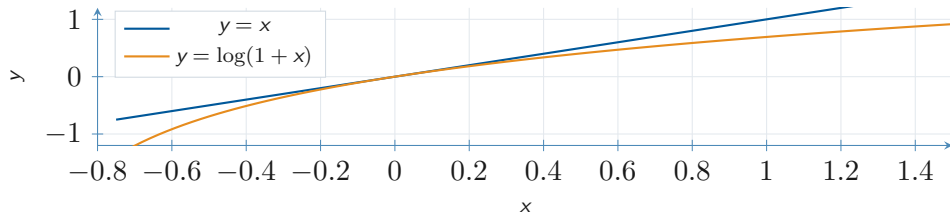
Technology Check: R Code

```
> prices <- c(100, 102, 101)
> arith_ret <- diff(prices) / head(prices, -1)
> log_ret <- diff(log(prices))
```

Key Definition / Result

$$u_t = \log \left(\frac{S_t}{S_{t-1}} \right) = \log(1 + R_t) \approx R_t$$

when R_t is small.



Pause-and-Think

Why does the approximation become worse when returns are large in magnitude?

Key Definition / Result

Multi-period log return adds:

$$\log \left(\frac{S_t}{S_0} \right) = \log \left(\frac{S_1}{S_0} \right) + \cdots + \log \left(\frac{S_t}{S_{t-1}} \right) = u_1 + \cdots + u_t$$

Deepening Layer

The implication is that multi-day risk can be linked to a **sum of daily shocks**. This connects naturally to Central Limit Theorem ideas.

Key Definition / Result

- ▶ **Average return rate**, or sample mean: $\bar{R} = \frac{1}{n} \sum_{t=1}^n R_t$.
- ▶ **Volatility**, or sample standard deviation: $s_R = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (R_t - \bar{R})^2}$.

Common Mistake / Pitfall

Volatility is **symmetric**: it treats big gains and big losses similarly. Risk managers care especially about **big losses**. This motivates VaR and CVaR.

Technology Check

Use R or Excel to compute the mean and standard deviation of returns.

Technology Check: R Code

```
> returns <- c(0.12, 0.16, 0.21, 0.15,  
-0.09, 0.55, 0.15, -0.03, 0.02,  
0.11, 0.38, 0.44)  
> mean(returns)  
> sd(returns)
```


Part C. Value-at-Risk

Key Definition / Result

VaR at probability p , such as $p = 0.05$, is the p -**lower quantile** of returns over a horizon T .

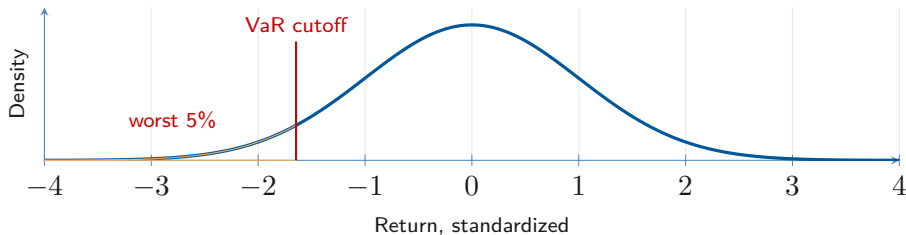
The confidence level is $c = 1 - p$. For example, $p = 0.05$ corresponds to a 95% confidence level.

Investigation / Activity: Interpretation Sentence

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'With 95% confidence, the return will not be worse than VaR.'

Visual: VaR Is A Cutoff In The Left Tail



Common Mistake / Pitfall

If “95% VaR = -25% ”, it does **not** mean you will lose 25%. It means only 5% of days are worse than -25% .

Key Definition / Result

Using **returns**, VaR is often negative because it represents a bad return. Using **loss** $L = -R$, VaR becomes a positive number.

Worked Example

If 95% VaR for returns is -0.25 , then 95% VaR for losses is $+0.25$.

Common Mistake / Pitfall

Always state which convention you use. Otherwise, a negative VaR and a positive VaR may appear to contradict each other even though they describe the same tail cutoff.

Key Definition / Result

Historical VaR uses the empirical distribution of past returns. It does not require a Normal distribution assumption.

1. Collect returns $R_1, \dots, R_n$.
2. Sort returns from worst to best.
3. Pick the p -quantile, such as the 5% cutoff.
4. Interpret the cutoff in context.

Worked Example

Data: 0.12, 0.16, 0.21, 0.15, -0.09 , 0.55, 0.15, -0.03 , 0.02, 0.11, 0.38, 0.44. Estimate 5% VaR.

1. Sort the 12 returns.
2. 5% of 12 is 0.6, so the cutoff is near the worst observation or observations.
3. With a small sample, interpret carefully. In practice, use larger n .

Technology Check

Use `quantile(returns, probs = 0.05, type = 7)` in R or `PERCENTILE.INC` in Excel.

Technology Check

Compute the empirical 5% and 1% quantiles directly.

Technology Check: R Code

```
> quantile(returns,  
probs = c(0.05, 0.01))
```

Key Definition / Result

Assume future returns are approximately $N(\bar{R}, s_R^2)$. Then the p -VaR is

$$\text{VaR}_p = \bar{R} + z_p s_R,$$

where z_p is the p -quantile of $N(0, 1)$, such as $z_{0.05} \approx -1.645$.

Common Mistake / Pitfall

This is a **model-based shortcut**. Always compare it with historical VaR when possible.

Investigation / Activity

Tutorial task, optional: derive the Normal-approximation VaR formula by standardising:

$$P(R < x) = p \Rightarrow P\left(\frac{R - \bar{R}}{s_R} < \frac{x - \bar{R}}{s_R}\right) = p.$$

Deepening Layer

Lecture focus: interpretation and workflow:

data $\rightarrow$ parameters $\rightarrow$ quantile cutoff $\rightarrow$ interpretation.

Worked Example

For the sample returns, suppose $\bar{R} = 0.1808$, $s_R = 0.1904$. Compute 95% and 99% VaR by Normal approximation.

1. 95% VaR uses $p = 0.05$, so $z_{0.05} \approx -1.645$.
2. 99% VaR uses $p = 0.01$, so $z_{0.01} \approx -2.326$.
3. Compute $\text{VaR}_{0.05} = \bar{R} - 1.645s_R$ and $\text{VaR}_{0.01} = \bar{R} - 2.326s_R$.

Key Definition / Result: Answer

$$\text{VaR}_{0.05} \approx -0.1324, \quad \text{VaR}_{0.01} \approx -0.2621.$$

Technology Check

Reproduce $\bar{R}$, s_R , VaR95, and VaR99 from the data.

Technology Check: R Code

```
> mu <- mean(returns)
> sig <- sd(returns)
> VaR95 <- mu + qnorm(0.05) * sig
> VaR99 <- mu + qnorm(0.01) * sig
> c(mu=mu, sig=sig, VaR95=VaR95, VaR99=VaR99)
```

Key Definition / Result

If returns are approximately independent and Normal, a common rule-of-thumb approximation is

$$\sigma_T \approx \sqrt{T} \sigma_1,$$

so

$$\text{VaR}_{p,T} \approx T\mu_1 + z_p\sqrt{T}\sigma_1.$$

Common Mistake / Pitfall

Square-root-of-time scaling is an approximation. It can fail when returns are not independent or when tails are heavy.

Pause-and-Think

Why might the worst week not be just $\sqrt{5}$ times the worst day?

Key Definition / Result

If you report 95% VaR, with $p = 0.05$, then about 5% of days should be **worse than VaR**. These days are called **exceptions**.

1. Compute daily VaR using a rolling window, such as the last 250 days.
2. Count how often the next-day return is below the VaR.
3. Compare the exception rate to p , which should be near 0.05 if the model is reasonable.

Part D. CVaR And Expected Shortfall

- ▶ VaR tells the cutoff, meaning the worst $p\%$ threshold.
- ▶ VaR does **not** tell how severe losses are beyond the cutoff.
- ▶ CVaR answers:

If we are in the worst $p\%$, what is the average loss?

Common Mistake / Pitfall

Two portfolios can have the **same VaR** but very different tail losses. CVaR distinguishes them.

Key Definition / Result

CVaR at probability p , using the returns view, is

$$\text{CVaR}_p = E[R \mid R \leq \text{VaR}_p].$$

Data version: average of the worst $p\%$ observations.

Technology Check

In tutorial, we compute CVaR from real data using sorting and averaging.

Key Definition / Result

Sample CVaR algorithm:

1. Sort returns: $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(n)}$.
2. Let $k = \lceil pn \rceil$. The tail set is $\{r_{(1)}, \dots, r_{(k)}\}$.
3. Compute $\widehat{\text{CVaR}}_p = \frac{1}{k} \sum_{i=1}^k r_{(i)}$.

Common Mistake / Pitfall

If n is small, k is tiny. Therefore, CVaR is unstable. Use larger samples whenever possible.

Worked Example

Suppose X is uniform on $[0, 100]$. Find 5% VaR and 5% CVaR, treating X as a loss.

1. 5% VaR is the cutoff so that 5% are below it: $\text{VaR}_{0.05} = 5$.
2. In the worst 5% region, losses are uniformly spread from 0 to 5.
3. The average of 0 to 5 is 2.5, so $\text{CVaR}_{0.05} = 2.5$.

Key Definition / Result: Answer

$$\text{VaR}_{0.05} = 5, \quad \text{CVaR}_{0.05} = 2.5.$$

Technology Check

Simulate many losses from the model, then compute empirical VaR and CVaR.

Technology Check: R Code

```
> set.seed(2)
> sim_loss <- runif(100000, min=0, max=100)
> VaR5 <- quantile(sim_loss, 0.05)
> CVaR5 <- mean(sim_loss[sim_loss <= VaR5])
```

Key Definition / Result

When an integral is hard, approximate the tail average by **probability-weighted boxes**:

$$E[R1(R \leq \text{VaR})] \approx \sum (\text{return}) \times (\text{probability mass}).$$

Pause-and-Think

Why does approximating an area by boxes make CVaR easier to understand?

Worked Example

Worst-case return table, tail only:

Probabilities: 0.5%, 1.5%, 2.5%, 3.5%, 4.5%, cumulative tail.

Returns: -50%, -39%, -25%, -20%, -10%.

Find 4.5% VaR and 5% CVaR, treating returns.

1. 4.5% VaR is the cutoff at cumulative probability 4.5%, so $\text{VaR} = -10\%$.
2. 5% CVaR is the average return in the worst 5% region.
3. Use probability-weighted average of the tail rows that sum to 5%.

Technology Check

Tutorial focus: do the weighted average carefully, including the small remaining probability slice.

Technology Check

Compute CVaR for real returns using sorting and averaging.

Technology Check: R Code

```
> VaR5 <- as.numeric(quantile(returns, 0.05))  
> CVaR5 <- mean(returns[returns <= VaR5])  
> VaR1 <- as.numeric(quantile(returns, 0.01))  
> CVaR1 <- mean(returns[returns <= VaR1])
```

	VaR	CVaR
Meaning	Tail cutoff , or quantile	Tail average , or severity
Question	"How bad is the worst $p\%$?"	"How bad on average in worst $p\%$?"
Strength	Simple and widely used	Tail-sensitive and conservative
Weakness	Ignores severity beyond cutoff	Needs more tail data for stability

Key Definition / Result

In risk communication, always report **both the number and its interpretation sentence**.

Part E. Investigation And Review

Technology Check

Tutorial focus: computing and communication, with technology-heavy workflows.

- ▶ Task A: Historical VaR and CVaR from a real return dataset.
- ▶ Task B: Normal-approximation VaR using estimates of $\bar{R}$, s_R , and z_p .
- ▶ Task C: Compare methods and write interpretation sentences.
- ▶ Optional proof: z-standardisation derivation for Normal VaR and CVaR.

Investigation / Activity

Activity. Pick one stock or ETF. Download daily closing prices.

1. Compute daily returns.
2. Compute 95% VaR by historical method and Normal approximation.
3. Compute 95% CVaR from data.
4. Write **one-sentence interpretations** for each number.
5. Discuss when the two VaR methods differ a lot.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Design a step-by-step R workflow to compute historical VaR, Normal-approximation VaR, and CVaR from daily closing prices.

1. Explain in words what a CDF tells you.
2. Given 10 numbers, describe how to compute the ECDF at $x = 0$.
3. What is a median from the quantile viewpoint?
4. Define arithmetic return R_t .
5. Why does volatility fail to distinguish upside from downside?
6. Interpret: “95% VaR = -0.10 ”, using the returns convention.
7. Describe the steps of historical VaR.
8. Describe CVaR in one sentence.

Investigation / Activity: Review Set XB

1. Two portfolios have identical 5% VaR. Give a scenario where CVaR differs.
2. Explain why log returns link to “sum of shocks”.
3. Compute VaR95 in a Normal model ($\mu = 0, \sigma = 0.02$).
4. For $n = 250$ returns, estimate 1% CVaR.
5. List two reasons Normal approximation fails.
6. Explain why CVaR is more conservative.

Key Definition / Result: Take-Home

- ▶ VaR = **quantile cutoff**.
- ▶ CVaR = **average tail loss**.
- ▶ Data + tech make it practical.

End Of Lecture 8